Wayne Hendrickson's Pioneering Discovery in Crystallography

A. Anchor Scene Candidates

1. "Farm Boy" Heads to College (Spring Valley, Wisconsin, 1959): In a small dairy farm town, 18-year-old Wayne Hendrickson planned to stay and help on his family farm until a fateful chat with his high school principal. The principal, a fellow Norwegian-American, sternly told him *"You vill go to college"*, nudging Hendrickson off the farm and into academia. Participants: Hendrickson and Principal. Setting: High school office in rural Wisconsin. What happened: A mentor figure intervened in Hendrickson's life path. Why it mattered: This candid moment launched Hendrickson's education, setting the stage for his future in science. *(Source: Oral history interview – first-person recollection.)*
2. An Unorthodox Path at NRL – "Off the Mainstream" (Washington, D.C., 1970s): After a PhD in biophysics at Johns Hopkins, Hendrickson took an unusual postdoc with crystallographer Jerome Karle at the Naval Research Lab, bypassing the usual elite route in Cambridge. Immersed in the esoteric "phase problem," he developed new computational methods (like the Hendrickson–Konnert refinement program) instead of chasing flashy biological structures. Participants: Hendrickson, Jerome Karle, NRL colleagues. Setting: Karle's lab, an insulated military research institute. What happened: Hendrickson focused on fundamental method development (phase determination, refinement) but *"hadn't accomplished... a big showy piece"* by postdoc's end. He struggled to land a professorship while peers from Cambridge easily secured top jobs. Why it mattered: This setback – *"I had made a bad career choice"* he admitted – forced Hendrickson to carve out a niche as a methods innovator. The "wilderness" years honed his resolve to solve the phase problem, even as he felt sidelined from mainstream academia. *(Source: Oral history – first-person, frank assessment.)*
3. Cracking the Phase Problem with Sulfur (NRL, 1981): In a marathon experiment that became legend, Hendrickson attempted to solve a protein structure *without any heavy-atom aids* – a bold test of anomalous diffraction. Over multiple days, he measured X-ray data from tiny crambin crystals, relying on the weak signal of sulfur atoms. Participants: Hendrickson (data collector), colleague Martha Teeter (collaborator on crambin). Setting: A basement lab at NRL, with a four-circle diffractometer and data output on paper tape. What happened: As Hendrickson laboriously collected intensities one reflection at a time, the punched paper tape recording the data unspooled across the floor. A janitor unknowingly ripped out yards of tape during cleanup – nearly trashing the experiment – but left it on the

floor. Undeterred, Hendrickson retrieved and processed the data, applying a new phase calculation method. Against the odds, he obtained a clear 1.5 Å electron density map using only the *anomalous scattering* from the protein's six sulfur atoms. Why it mattered: This was a eureka moment – the first structure (crambin) solved by single-wavelength anomalous diffraction (SAD) from native sulfur. It *"was the strongest proof of the impact of anomalous scattering,"* Hendrickson later said, earning him the ACA's inaugural Patterson Award. *(Sources: First-person interview; commentary by J. L. Smith – detailed, well-sourced account.)*

4. MAD Breakthrough – Streptavidin at the Synchrotron (Brookhaven Lab, 1988): Now a Columbia University professor (and newly an HHMI investigator), Hendrickson pushed his phasing quest further by exploiting tunable synchrotron X-rays. On beamline X4A at Brookhaven, he and colleagues (including postdoc J. L. Smith) soaked crystals of streptavidin with a selenium-containing biotin derivative. Participants: Hendrickson's team (Pähler, Smith, Satow, Merritt) and beamline scientists (Phizackerley, Sweet). Setting: NSLS beamline, collecting data at three carefully chosen X-ray wavelengths. What happened: Using the multi-wavelength anomalous diffraction (MAD) method that Hendrickson had formulated, the team measured selenium's anomalous signals at different wavelengths from a single crystal. They successfully computed phases de novo. Why it mattered: In 1989, Hendrickson published the 3.1 Å crystal structure of core streptavidin – the first protein structure determined entirely by MAD phasing. The paper proclaimed the effectiveness of MAD *"for macromolecular crystallography"*, validating MAD as a powerful general technique. This achievement "established [MAD] as the method of choice" for protein structure determination, paving the way for high-throughput structural genomics. *(Sources: Primary journal article – PNAS 1989 (co-authored by Hendrickson); Gairdner Foundation citation – high-quality secondary.)*
5. HIV's Lock-and-Key Imaged – CD4 Structure Race (New York, 1990): In the midst of the AIDS crisis, Hendrickson collaborated with Columbia colleague Richard Axel to crystallize CD4, the T-cell receptor that HIV uses to enter cells. Another team (led by J. H. Wang) was working in parallel. Participants: Hendrickson's group (incl. postdoc S. E. Ryu) and Axel; competing lab in Boston (Wang et al.). Setting: Columbia lab and National Synchrotron, late nights solving a structure with heavy social urgency. What happened: Both groups succeeded in determining the domain structure of CD4 almost simultaneously. Hendrickson's team solved a two-domain fragment of CD4 at 2.3 Å, using MAD phasing with a recombinant construct. The results were published back-to-back in *Nature* in November 1990. Why it mattered: These first molecular images of CD4 revealed how HIV's gp120 might bind, fueling vaccine design efforts. The achievement

was *front-page news in The New York Times*. Hendrickson recalls an AIDS activist phoning him after seeing the headline, urgently suggesting an artificial sweetener (aspartame) might block HIV by fitting into CD4's binding pocket. *(Sources: Hendrickson interview – first-person context; Science press summary – notes dual Nature papers.)*

6. Structural Biology's Crown Jewels – Kinase and HIV Envelope (1994–1998): Hendrickson's methods unlocked ever more complex and medically crucial proteins. In 1994, his group (with postdoc S. R. Hubbard) solved the first tyrosine kinase structure – the insulin receptor's kinase domain – by MAD at 2.1 Å. This revealed the kinase's auto-inhibition mechanism, a paradigm for targeted cancer drug design. A few years later, Hendrickson partnered with NIH researchers to determine the structure of HIV's gp120 envelope glycoprotein bound to CD4 and a neutralizing antibody. Participants: For the kinase: Hendrickson, Hubbard, Lily Wei, Lou Ellis; for HIV gp120: Hendrickson, Peter Kwong (grad student), and collaborators at NIH. Setting: Synchrotron beamlines and Hendrickson's lab, mid-90s. What happened: Both structures were technically demanding – the kinase required capturing multiple phosphorylation states, and gp120 required stabilizing a glycoprotein complex – but MAD phasing and selenomethionine labeling (another Hendrickson innovation) made the solutions feasible. Why it mattered: Hendrickson's work provided atomic blueprints for two central biomedical targets: tyrosine kinases (leading to drugs like Gleevec) and the HIV entry mechanism (guiding vaccine antibody design). These high-impact successes cemented MAD's reputation and showed structural biology's direct relevance to medicine. *(Sources: Journal abstracts – Nature 1994 (insulin receptor); Gairdner Foundation profile.)*

B. Direct Quotes from Wayne Hendrickson

- *"Anomalous scattering. That's a big theme of my life."* (Emphasizing how solving the phase problem via anomalous diffraction became his lifelong pursuit.)
- *"I had made a bad career choice… The standard thing was... you went off to the Lab of Molecular Biology in Cambridge… I hadn't been blessed properly… I didn't have a big showy piece to show at the end."* (Reflecting on his unconventional postdoc path and why he initially struggled to get a job.)
- *"It was a huge sacrifice that I made going there [NRL]… I was so focused on this very basic question of the phase problem… How do you even explain that to your mother?"* (On dedicating himself to an abstract problem that was hard to explain outside the crystallography community.)
- *"We solved the structure of CD4 and published it, and it received some attention. It was on the front page of The New York Times."* (Underscoring the public impact of his lab's work on the HIV-related CD4 protein.)

- *(On an AIDS activist's idea after seeing the CD4 structure) "And I said, 'Wow. [laugh] … I guess your work has some relevance, Wayne.' But the problem is, the guy is impatient… these people were desperate."* (Recalling an outside phone call that pressed him for an immediate AIDS cure, and his realization of the urgent expectations facing scientists.)
- *"The one [award] that was really fun for me was recognition from the American Crystallographic Association as the first recipient of the Patterson Award… I felt really proud that I was, in some ways, a scientific descendant of Patterson… recognizing the things we had done on this crambin structure with anomalous scattering."* (Expressing personal pride in receiving an award for the anomalous-diffraction solution of crambin, connecting him to crystallography pioneer A. L. Patterson.)

C. Human Reality Details

- Humble farm origins: Hendrickson grew up as the eldest of seven children on a Wisconsin dairy farm, doing daily barn chores ("*my sisters did the dishes, and we did the cows*," he quipped). His parents, children of Norwegian immigrants with little formal education, supported his schooling even though college was outside their experience.
- Curiosity from nature: As a boy, he roamed the woods and built his own camera tripod to photograph birds in their nests. This self-driven curiosity about the natural world "sparked a curiosity in nature that never ended" and primed him for scientific inquiry.
- Homespun intervention: Hendrickson's matter-of-fact decision to stay on the farm was upended by his high school principal's intervention. In a thick Norwegian accent the principal insisted on college, an encounter Hendrickson recalls with humor and gratitude. This pivotal nudge gave a first-generation student the confidence to pursue science.
- Academic isolation and resilience: At NRL, Hendrickson felt "in a cocoon" – financially supported but outside academia's mainstream. He jokingly notes the Navy refused to let him spend an NIH grant he won, which frustrated him but also *"motivated me to leave"* for a university position. Cut off from a typical tenure-track environment, he nevertheless became deeply known in the crystallography circle through conference participation and method innovations.
- Tape on the floor: The crambin experiment became a lore-filled episode illustrating Hendrickson's tenacity. Over days of data collection, punched paper tape containing the X-ray intensity data piled up on the floor until a janitor inadvertently tore it out. Fortunately, the precious tape wasn't thrown away – Hendrickson recovered it and managed to compute the structure. This

near-disaster, narrowly averted by perseverance and luck, underscores the very human challenges behind scientific breakthroughs.

- Activist pressure: Solving HIV-related structures in the 1980s put Hendrickson in the public eye. After his CD4 structure hit headlines, activists fighting for their lives sought him out with ideas. One caller hypothesized an artificial sweetener might block HIV, a sign of desperate hope. Hendrickson empathized but recognized his limits: *"I really concluded I don't have the right background to do the medical application work… it would be smarter to continue very basic work"*, allowing others to build on his findings in the clinic. This insight showed his self-awareness to stick to what he does best.
- Lab culture of collaboration: Upon moving to Columbia in 1984, Hendrickson built a highly collaborative lab. He "made alliances" on diverse protein projects as test-beds for new technology. By working on partners' biological problems (ion channels, receptors, viral proteins), his team advanced methods while co-discovering important structures – a win-win approach that became a hallmark of his mentoring style.
- Mentoring and lineage: Hendrickson's influence is reflected in the success of his trainees. For example, Peter Kwong, who co-authored the landmark gp120 structure as a PhD student, went on to lead vaccine design at NIH. Hendrickson himself was mentored by top crystallographers (his PhD advisor Warner Love and postdoc mentor Jerome Karle), but he notes that it was the *field* of crystallography that truly attracted him, more so than chasing famous mentors. This independent streak shaped his lab's ethos of pursuing fundamental problems over fashionable topics.

D. Science Connection (Intro Biochemistry Context)

Topic – Protein Structure Determination & Phase Problem: In X-ray crystallography, determining a protein's 3D structure requires not only collecting diffraction patterns, but also recovering the "phase" information for each diffracted wave. This infamous phase problem is like having a complex jigsaw puzzle with pieces' shapes (intensities) but missing crucial orientation clues (phases) needed to reconstruct the picture. Traditional solutions involved adding heavy atoms to crystals and comparing datasets, or guessing parts of the structure, to approximate phases. Hendrickson's Contribution – MAD Phasing: Wayne Hendrickson pioneered Multi-Wavelength Anomalous Diffraction (MAD) as a direct way to solve phases. By incorporating an X-ray-scattering element (often selenium by substituting selenomethionine in the protein) and collecting data at several X-ray wavelengths around that element's absorption edge, crystallographers can tease out phase information from a single crystal. It's analogous to taking multiple flash photos through different colored filters – the subtle differences reveal hidden structural details. Hendrickson's MAD method provided an "experimental shortcut" to compute phases

accurately, eliminating the need for multiple heavy-atom derivatives. Bridge to Biochemistry: For undergraduates, think of protein crystallography like trying to reconstruct a song when you can hear its volume but not the tune. The phase problem is losing the tune. Hendrickson gave scientists a clever way to recover the tune by exploiting the slight off-beat "anomalous" signals from specific atoms. His MAD phasing innovation transformed structural biology: suddenly, determining new protein structures became faster and more routine. This opened the door to structural genomics – solving structures en masse – and provided molecular blueprints for enzymes, receptors, and viral proteins at an unprecedented rate. In short, Hendrickson enabled biochemists to *see* what proteins really look like, which is fundamental to understanding their function and designing drugs. *(Analogy source: general crystallography principle – not a direct quote; Contribution and impact source: Gairdner Award citation.)*

E. Complications/Controversies

Overall, Hendrickson's work avoided major scientific controversies – his contributions were quickly embraced by the crystallographic community. No serious disputes over credit appear in the record; in fact, he shared the 2004 Paul Janssen Prize with another phasing-method pioneer (M. G. Rossmann), reflecting collaborative recognition. One notable *competition* occurred in 1990 when two groups solved the CD4 structure concurrently. Hendrickson's team (with S. E. Ryu) and a Harvard team (J. Wang et al.) published back-to-back papers in *Nature*, each revealing the CD4 receptor's fold. While essentially a tie, it underscored the intense race to illuminate HIV's biology rather than a personal conflict. Hendrickson's account shows no animosity – he collaborated with immunologist Richard Axel on CD4 and acknowledged the media spotlight that followed. If anything, Hendrickson's career complication was being ahead of his time. His fixation on solving the phase problem was initially seen as *"off the mainstream"*, making it harder to secure a top academic job in the late 1970s. He humorously noted that not having the Cambridge pedigree meant he "hadn't been blessed properly" by the establishment. This could have derailed his career, but he turned it into an asset: by the time MAD proved revolutionary, any early doubts about his path were moot. Colleagues universally recognized his method's value (he received the Aminoff Prize, Ewald Prize, etc., with no hint of controversy). In interviews and writings, Hendrickson comes across as modest and collaboration-oriented, focusing on solving problems rather than on personal credit. Our research uncovered no evidence of priority disputes or ethical conflicts attached to his name. *(Sources: Nature references for concurrent CD4 papers; Hendrickson interview on mainstream vs. offbeat career; Columbia/BNL accolades.)*

F. "Not on Wikipedia" Findings

(Distinctive insights from primary sources that go beyond Hendrickson's Wikipedia entry:)

- Farm-boy to physicist: Hendrickson was raised on a rural Wisconsin farm by Norwegian-immigrant parents. Until age 17 he expected to be a farmer; he even won prizes in high school agriculture classes. His principal's blunt directive to attend college changed that trajectory. (His Wikipedia page mentions nothing of this formative farm upbringing or the principal's influence.)
- DIY naturalist: As a teenager, Hendrickson displayed inventive curiosity – he built a camera tripod from scratch to photograph wildlife around his farm, capturing images of birds in their nests. This early tinkering with optics and nature is not noted on Wikipedia, yet it foreshadows his experimental ingenuity in later life.
- Punched-tape near disaster: During the breakthrough crambin experiment, Hendrickson's data were recorded on paper tape. A cleaning staff member accidentally ripped out the long tape roll mid-experiment. In a stroke of luck, the tape remained in the lab for Hendrickson to find on the floor and salvage the data. This almost comic mishap – pivotal to a landmark discovery – is absent from Wikipedia's account of his work.
- "We seldom take Americans": As a young man, Hendrickson actually applied to Francis Crick's biophysics program at Cambridge. He received a rejection letter stating they *"seldom, if ever, take Americans"* (Crick had just had one American student). This little anecdote of transatlantic elitism, and how Hendrickson ended up at Hopkins instead, isn't on Wikipedia but provides insight into the twists of his early career and perhaps his later outsider perspective.
- Activist phone call: After Hendrickson's lab solved the CD4 structure, he got an unsolicited call from an AIDS activist who read about it. The caller theorized that the sweetener aspartame (phenylalanine being a component) might block HIV entry by binding CD4's pocket. Wikipedia doesn't mention this encounter, which humanizes the era's desperation and how Hendrickson's work intersected with real-world hope and pressure.
- Founding a company and journals: Beyond academia, Hendrickson co-founded a biotech startup called SGX Pharmaceuticals (focused on structural genomics-based drug discovery) and was a founding editor of key journals *Structure* and *Current Opinion in Structural Biology*. His Wikipedia page lists many honors but omits these entrepreneurial and editorial roles that highlight his broader influence on the field.
- Hendrickson–Lattman coefficients: While a graduate student, Hendrickson (with Ed Lattman) developed the probability representation for crystallographic phases known as H-L coefficients. This 1970 innovation is fundamental in phasing calculations, yet Wikipedia's biography of Hendrickson does not mention it. It shows that even early in his career, he was addressing the phase problem – a through-line to his later MAD work.

(Sources: Oral history interview for farm, principal, naturalist details; IUCrJ commentary for tape incident and H-L coefficients; Oral history for Cambridge rejection and activist call; BNL news for SGX and journal founding.)

G. Sources Consulted

1. Oral History – Wayne Hendrickson (2020): Transcript of an interview by David Zierler on April 13, 2020. Hosted by the American Institute of Physics (AIP) and American Crystallographic Association (ACA), this first-person source provides extensive insight into Hendrickson's childhood, education, career decisions, scientific philosophy, and anecdotes (e.g. farm life, NRL years, HIV research). *Quality:* Primary source (firsthand quotes and memories).
2. Gairdner Foundation Profile (2003): Award citation for Wayne A. Hendrickson's 2003 Canada Gairdner International Award. Summarizes his contributions to macromolecular crystallography, including development of MAD phasing and key structures solved (tyrosine kinase, HIV gp120). *Quality:* Secondary, authoritative (award committee summary).
3. Janet L. Smith, "Commentary on 'Facing the phase problem'" (IUCrJ, 2023): A commentary on Hendrickson's Ewald Prize monograph, appearing in *IUCrJ* 10(5):555-556. It recounts the history of phase determination methods with specific focus on Hendrickson's work. Notably details the crambin sulfur SAD experiment (including the punched tape story) and the development of Hendrickson–Lattman coefficients and restrained refinement with Konnert. *Quality:* Secondary, expert (written by a fellow crystallographer), with historical details drawn from first-hand knowledge.
4. Brookhaven National Lab News (Apr 30, 2009): Press release titled "Hendrickson Appointed NSLS-II Associate Project Director for Life Sciences". Provides a concise biography, highlighting his education, positions, and technical achievements (stereochemical restrained refinement, MAD method, selenomethionine protein production, beamline leadership). Also notes his roles in founding journals and a biotech company. *Quality:* Secondary, institutional (fact-checked career summary).
5. Hendrickson et al., PNAS 86:2190-2194 (1989): Research article "Crystal structure of core streptavidin determined from multiwavelength anomalous diffraction…". This is the seminal paper demonstrating MAD phasing on a protein crystal. The abstract explicitly states the structure was analyzed from diffraction data at three wavelengths and that the study *"demonstrates the effectiveness of [MAD] procedures"*. *Quality:* Primary, peer-reviewed.
6. Hubbard et al., Nature 372:746-754 (1994): Research article "Crystal structure of the tyrosine kinase domain of the human insulin receptor". Solved by MAD phasing and refined to 2.1 Å, this paper (co-authored by Hendrickson) is cited for

content on using MAD to determine a biologically important structure and revealing its mechanism. *Quality:* Primary, peer-reviewed.

7. Nature Milestone "Anomalous diffraction tackles phasing" (Nature 511, p.17, 2014): A short retrospective article by K. R. Legate describing the development of anomalous scattering methods. Highlights Hendrickson & Teeter's 1981 sulfur-anomalous solution of crambin and the concept of multi-wavelength phasing (cites Hendrickson 1985 ACA Transactions paper). *Quality:* Secondary, journal news piece.
8. "The Immunology Connection – my first T cell receptor structure projects" (2014, OUP blog excerpt): Personal recollection by immunologist Ellis Reinherz (Harvard) – mentions simultaneous 1990 Nature papers on CD4 by two labs and the coverage in NYTimes. Used to confirm the competitive context of the CD4 structure determination. *Quality:* Secondary, memoir-style (provides context on timeline).
9. Additional references: Hendrickson's Wikipedia entry was consulted for an overview of his accomplishments (to ensure our "not on Wikipedia" finds were novel). PubMed entries for the 1998 HIV gp120 structure (Kwong *et al.*, *Nature* 393:648–659) and other works were used to verify participants and outcomes. No information was taken solely from Wikipedia, and all unique details were cross-checked with primary sources.

(All references above were accessed and verified in October 2023.)

H. Lens Recommendation

Lens Scoring (0–3):

- Eureka (Discovery crystallized): 3 – Hendrickson's career offers well-documented "aha" moments with rich details (e.g. the sulfur SAD success with crambin and the multi-wavelength breakthrough on streptavidin). These scenes are vividly recounted with first-hand sourcing, making a discovery-focused narrative compelling.
- Failure (Setback shapes trajectory): 3 – We have clear evidence of a pivotal setback (his "bad career choice" of not going to LMB, leading to job rejections) that forced him down a different path. That struggle directly set the stage for his innovative work at NRL, a classic case of overcoming adversity fueling success.
- Mentor (Mentorship pivot): 2 – There are mentor figures (a high school principal's push, Jerome Karle's patronage at NRL), but no single mentor–student interaction defines his entire trajectory. The principal's influence is compelling but very early, and later Hendrickson was more self-driven by the "field" than guided by a superstar mentor.

- Controversy (Conflict/competition): 1 – Aside from a friendly race to solve CD4, Hendrickson's story lacks major conflicts. He wasn't embroiled in public disputes; the narrative has more internal tension (mainstream vs. maverick) than interpersonal controversy.
- Human Detail (Personal life illuminates science): 3 – His life furnishes striking personal details that connect to his science: a farm-born tinkerer turned structural biologist, enduring isolation at NRL to crack a fundamental problem, and humorous, humanizing anecdotes like the tape incident and activist call. These well-sourced details would enrich any narrative and tie directly into his scientific motivations.

Recommended Primary Lens – *Eureka*: Hendrickson's story is most powerfully told as a series of scientific breakthrough moments emerging from perseverance. The Eureka lens fits best because the evidence centers on dramatic discoveries: for instance, scene #3 (the 1981 crambin experiment) and scene #4 (1988–89 MAD phasing triumph) are climactic, scene-driven breakthroughs. These moments literally changed how structural biologists solve structures, which provides a thrilling narrative spine. Hendrickson's "phase problem" obsession builds to tangible eureka payoffs, giving the narrative a natural arc of problem→insight→resolution. Moreover, our richest anecdotes (the janitor tearing the tape, the one-crystal MAD proof-of-concept) support a discovery theme with concrete, suspenseful detail. Opening Scene: I suggest starting with Scene 3, the crambin data collection saga. It's vivid and symbolic – a lone scientist in a lab at 3 am, surrounded by loops of paper tape, on the verge of a breakthrough. This scene drops readers right into the tension and triumph of discovery. An early quote to consider is Hendrickson's own reflection, *"Anomalous scattering… that's a big theme of my life"*, which could appear as he puzzles over the crambin data. It foreshadows that this moment is the crystallization of his life's theme – solving the unsolvable. Backup Lens – *Failure to Fortune*: If not eureka, a close second is the Failure (underdog) lens (score tied at 3). Hendrickson's journey can be cast as an against-the-odds story – the farm boy not groomed for science, the young researcher dismissed by elites, working in obscurity yet ultimately reshaping his field. Scene #2 (mid-1970s, Hendrickson at NRL feeling passed over) would be a poignant opener under this lens. We'd see him at a low point – *"I didn't get jobs… I hadn't been blessed"* – then follow how that apparent "failure" set the stage for his innovative successes (Scenes 3 and 4 as the fruits of his outsider path). This approach highlights resilience and could engage readers emotionally with the personal stakes. The tone would celebrate how setbacks (lack of mentorship, institutional support) incubated a groundbreaking idea. An early quote for this angle might be his frank admission, *"I had made a bad career choice… [I] didn't have a big showy piece"*, to immediately establish the obstacle he had to overcome.

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